HU Collection
ESAT Mock Set F: Mathematics 1

Worked Solutions

Kevin Hu 29 October 2026

Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	A	D	B	D	B	A	E	A	C	A

Q	11	12	13	14	15	16	17	18	19	20
Ans	C	D	B	E	C	F	F	B	A	D

Q	21	22	23	24	25	26	27			
Ans	E	C	B	F	C	F	E			

Each solution gives an intended route and names the trap the question was built around. Diagrams show the configuration each solution works with.

Question 1. Answer **A**.

The curved surface of a hemisphere has area $2\pi r^2 = 18\pi$ m^2 $= 18\pi \times 10^4$ cm^2, and the thickness is 0.125×10^{-4} cm $= 1.25 \times 10^{-5}$ cm. The volume of gold is

$$18\pi \times 10^4 \times 1.25 \times 10^{-5} = 2.25\pi \text{ cm}^3,$$

so the mass is $2.25\pi \times 19.2 = 43.2\pi$ g, about 136 g.

Using πr^2, $4\pi r^2$ or $3\pi r^2$ (which wrongly includes the flat base) for the area gives 21.6π, 86.4π or 64.8π. The options 4.32π and 432π come from converting m^2 to cm^2 with a factor of 10^3 or 10^5 instead of 10^4.

Question 2. Answer **D**.

Each of the first two steps dilutes by a factor of 100 (the 1 ml ends up spread through 100 ml, not 99 ml), and the last step by $\dfrac{5}{0.5} = 10$. So the final mixture is 10^5 times weaker than the culture.

The final mixture contains about $\dfrac{63}{0.05} = 1260$ bacteria per ml, so the culture contains about $1260 \times 10^5 = 1.26 \times 10^8$ per ml, which is 1.26×10^{11} per litre.

Stopping at the value per ml gives 1.26×10^8, and multiplying by 0.05 instead of dividing gives 3.15×10^8.

Question 3. Answer **B**.

Take the total mass to be 1296 kg (any common multiple of 16 and 81 will do). Then there are 16 large animals and 81 small ones, and with k the constant the totals are

$$81 \times k \cdot 16^{3/4} = 81 \times 8k = 648k \qquad \text{and} \qquad 16 \times k \cdot 81^{3/4} = 16 \times 27k = 432k,$$

in the ratio $648 : 432 = 3 : 2$.

In general a herd of total mass T made of animals of mass M needs $\dfrac{T}{M} \times kM^{3/4} = kTM^{-1/4}$, so the ratio is $\left(\dfrac{81}{16}\right)^{1/4} = \dfrac{3}{2}$: for the same total mass, smaller animals need more energy. The ratio $27 : 8$

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compares one large animal with one small one, and $1:1$ treats the requirement as proportional to mass.

Question 4. Answer **D**.

Take Q's starting point as the origin, with x east and y north. After t hours, P is at $(3t,\ 10)$ and Q is at $(0,\ 4t)$, so the square of the distance between them is

$$9t^2 + (10 - 4t)^2 = 25t^2 - 80t + 100 = 25(t - 1.6)^2 + 36.$$

This is least when $t = 1.6$, that is after 1 hour 36 minutes, at 13:36, and the least distance is $\sqrt{36} = 6$ km.

The options at 13:00 and 14:00 give the correct separations at those times, which are not the least; 14:30 is when Q crosses P's path, which is not when they are closest.

Question 5. Answer **B**.

At a point at distances d_A and d_B from the loudspeakers, the intensities are equal when $\dfrac{1}{d_A^2} = \dfrac{4}{d_B^2}$, that is when $d_B = 2d_A$.

Between A and B, $d_A + d_B = 6$ gives $d_A = 2$. Beyond B, $d_A = d_B + 6$ is larger than d_B, whereas we need $d_B = 2d_A$, so there is no such point there. Beyond A, $d_B = d_A + 6 = 2d_A$ gives $d_A = 6$. So there are two points, 2 m from A towards B and 6 m from A on the side away from B, and they are 8 m apart.

Looking only between the loudspeakers finds just one point, and the option 2 m is its distance from A.

Question 6. Answer **A**.

After the extra matches are added, there are $200 + k$ matches, of which $80 + k$ have at most 1 home goal and $30 + k$ have none.

If $200 + k$ is odd, the median is the $\frac{201+k}{2}$th value, which is 1 when $30 + k < \frac{201+k}{2} \le 80 + k$; the right-hand inequality gives $k \ge 41$. If $200 + k$ is even, the median is the mean of the $\frac{200+k}{2}$th and $\frac{202+k}{2}$th values, and it is 1 only if both are 1, which needs $\frac{202+k}{2} \le 80 + k$, that is $k \ge 42$.

So the smallest value is $k = 41$, when the 121st of 241 values is the last 1. The trap is $k = 40$: then exactly half of the 240 values are at most 1, so the 120th value is 1 but the 121st is 2, and the median is 1.5.

Question 7. Answer **E**.

In centimetres, the plank length L satisfies $315 \le L < 325$ and the shelf length s satisfies $39.5 \le s < 40.5$. The number of shelves is the largest whole number n with $ns \le L$.

In the worst case L is as short and s as long as possible. Since $\dfrac{L}{s} > \dfrac{315}{40.5} = 7.7\ldots$, seven shelves always fit, but eight need not: for example $8 \times 40.4 = 323.2 > 315$. In the best case, $\dfrac{L}{s} < \dfrac{325}{39.5} = 8.2\ldots$, so nine shelves never fit, while eight can (for example when $L = 320$ and $s = 40$). Hence $G = 7$ and $H = 8$.

Dividing the stated values, $320 \div 40 = 8$, suggests $G = 8$, but this ignores the possibility that both measurements are in error in the unfavourable direction.

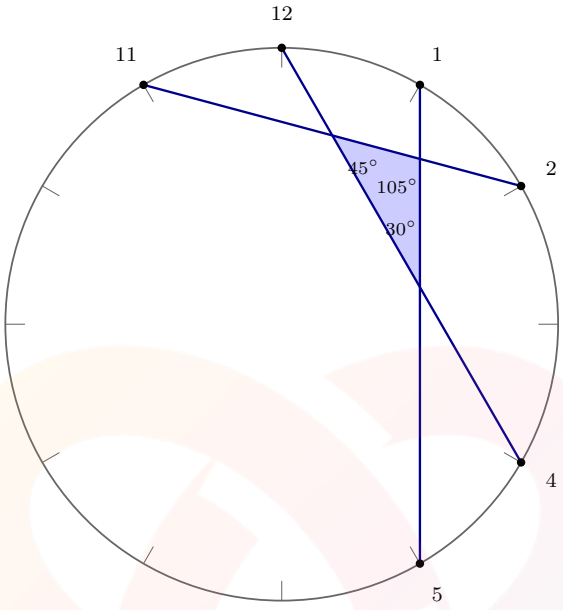
Question 8. Answer **A**.

Neighbouring hour marks cut off arcs of $30°$ at the centre. When two chords cross at a point inside a circle, each angle between them equals half the sum of the arcs cut off by that angle and by its vertically opposite angle. (Join two of the endpoints: the angle is an exterior angle of a triangle whose other two angles are inscribed angles standing on those two arcs.)

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The chords from 12 to 4 and from 1 to 5 make angles of $\frac{1}{2}(30^\circ + 30^\circ) = 30^\circ$ and 150°. The chords from 12 to 4 and from 2 to 11 make angles of $\frac{1}{2}(60^\circ + 30^\circ) = 45^\circ$ and 135°. The chords from 1 to 5 and from 2 to 11 make angles of $\frac{1}{2}(90^\circ + 60^\circ) = 75^\circ$ and 105°.

The triangle has one angle from each pair, and they must add up to 180°. The only possibility is $30^\circ + 45^\circ + 105^\circ$, so the largest angle is 105°, not the 75° that a quick look at the arcs between 1, 2, 5 and 11 might suggest.



Question 9. Answer **C**.

There are $\binom{36}{2} = 630$ pairs. Two dots that are a columns and b rows apart, with $0 \le a, b \le 5$, are $\sqrt{a^2 + b^2}$ apart, and this is a whole number only when $a = 0$, or $b = 0$, or $\{a, b\} = \{3, 4\}$.

There are $6 \times \binom{6}{2} = 90$ pairs in the same row and 90 in the same column. For $a = 3$ and $b = 4$, the left-hand dot can be in any of 3 columns, the lower dot in any of 2 rows, and the segment can slope up or down, giving $3 \times 2 \times 2 = 12$ pairs; similarly 12 pairs have $a = 4$ and $b = 3$.

The probability is $\dfrac{180 + 24}{630} = \dfrac{34}{105}$. Forgetting the 3, 4, 5 triangles gives $\dfrac{2}{7}$, and counting them in only one orientation gives $\dfrac{32}{105}$.

Question 10. Answer **A**.

The range is multiplied by $|a|$, so $|a| = \frac{50}{40} = 1.25$, and a may be 1.25 or -1.25. The mean becomes $45a + b$, which must be 60.

If $a = 1.25$ then $b = 3.75$: the median becomes $1.25 \times 47 + 3.75 = 62.5$ and the quartiles become 51.25 and 71.25.

If $a = -1.25$ then $b = 116.25$, and the order of the data is reversed. The median becomes $116.25 - 58.75 = 57.5$, and the old upper quartile turns into the new lower quartile, $116.25 - 1.25 \times 54 = 48.75$, with new upper quartile $116.25 - 1.25 \times 38 = 68.75$.

So I is true, II is true (the interquartile range is $1.25 \times 16 = 20$ in both cases), and III is false. Forgetting the negative value of a makes III look true and I look false.

Question 11. Answer **C**.

Over the 2 hours between doses the concentration is multiplied by $0.8^2 = 0.64$. If the concentration just after a dose settles to S, then 2 hours later it is $0.64S$, and the next dose takes it back to

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$0.64S + 18$, which must equal S. So $0.36S = 18$, giving $S = 50$, and the level just before each dose is $0.64 \times 50 = 32$.

The levels do approach this: just after the doses they are 18, 29.52, 36.89…, and the gap $50 - S_n$ is multiplied by 0.64 each time. The option 50 is the level just after a dose; treating the 2-hour fall as 40% gives 45 and 27, and using a factor of 0.8 per dose gives 90 and 72.

Question 12. Answer **D**.

Consecutive moving averages share three readings, so their difference is a quarter of the difference between the reading that enters and the reading that leaves:

$$T_{k+4} - T_k = 4 \times (\text{next average} - \text{this average}).$$

The first average gives $T_4 = 48 - 9 - 11 - 14 = 14$. Then $T_5 = 9 + 4 = 13$, $T_6 = 11 + 8 = 19$, $T_7 = 14 - 4 = 10$, $T_8 = 14 + 8 = 22$, $T_9 = 13 + 8 = 21$ and $T_{10} = 19 - 4 = 15$.

In order, the readings are $9, 10, 11, 13, 14, 14, 15, 19, 21, 22$, so the median is 14. The median of the seven moving averages is 15, which is a different thing.

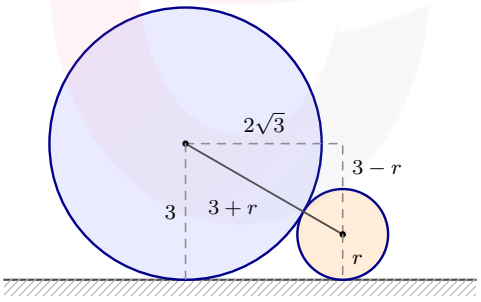
Question 13. Answer **B**.

Seen from above, the centres of the large balls form an equilateral triangle of side 6, and by symmetry the centre of the small ball is directly above the centre of this triangle, at a horizontal distance of $\dfrac{6}{\sqrt{3}} = 2\sqrt{3}$ from each large centre.

The large centres are 3 above the table and the small centre is r above it, and the centres of two touching balls are $3 + r$ apart. Pythagoras in the vertical plane through the two centres gives

$$(2\sqrt{3})^2 + (3 - r)^2 = (3 + r)^2,$$

so $12 = 12r$ and $r = 1$.



Treating the problem as one about circles in a plane gives $2\sqrt{3} - 3$, which ignores the different heights of the centres.

Question 14. Answer **E**.

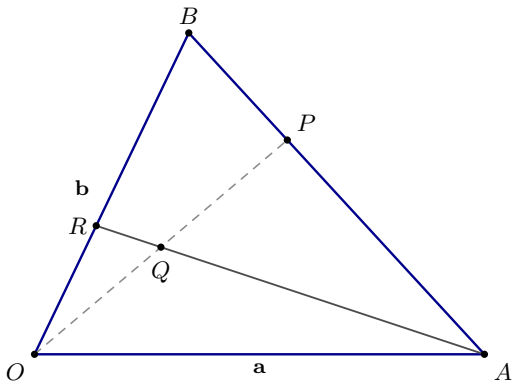
Let $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. Then $\overrightarrow{OP} = \mathbf{a} + \frac{2}{3}(\mathbf{b} - \mathbf{a}) = \frac{1}{3}\mathbf{a} + \frac{2}{3}\mathbf{b}$, so $\overrightarrow{OQ} = \frac{1}{6}\mathbf{a} + \frac{1}{3}\mathbf{b}$.

The points of the line AQ have position vectors

$$\mathbf{a} + t\left(\overrightarrow{OQ} - \mathbf{a}\right) = \left(1 - \tfrac{5}{6}t\right)\mathbf{a} + \tfrac{1}{3}t\,\mathbf{b}.$$

Such a point lies on OB when the coefficient of $\mathbf{a}$ is 0, that is when $t = \frac{6}{5}$, and then $\overrightarrow{OR} = \frac{2}{5}\mathbf{b}$. Hence $OR : RB = 2 : 3$.

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Question 15. Answer **C**.

The units of σ are kg s^{-2}, so the right-hand side has units

$$\mathrm{kg}^{a}\,\mathrm{m}^{-3a} \times \mathrm{m}^{b} \times \mathrm{kg}^{c}\,\mathrm{s}^{-2c} = \mathrm{kg}^{a+c}\,\mathrm{m}^{b-3a}\,\mathrm{s}^{-2c},$$

which must be seconds. So $-2c = 1$, $a + c = 0$ and $b = 3a$, giving $c = -\frac{1}{2}$, $a = \frac{1}{2}$ and $b = \frac{3}{2}$, that is $T = k\sqrt{\dfrac{\rho r^3}{\sigma}}$.

With ρ and σ unchanged, $T \propto r^{3/2}$. The radius is divided by 4, so T is divided by $4^{3/2} = 8$, giving 1 ms. Assuming $T \propto r$ gives 2 ms, and $T \propto r^2$ gives 0.5 ms.

Question 16. Answer **F**.

Count by the number of 50p coins. With two of them there is 1 way. With none, the sequences of 10p and 20p coins making £1 follow the Fibonacci pattern (a sequence for a total of n tens ends in a 10p after a total of $n-1$, or a 20p after a total of $n-2$), giving $1, 2, 3, 5, 8, \ldots$ for totals of $10\text{p}, 20\text{p}, 30\text{p}, \ldots$, and so 89 for £1.

With exactly one 50p, the other coins make 50p: either five 10p coins, three 10p and one 20p, or one 10p and two 20p. Including the 50p, these can be arranged in

$$\frac{6!}{5!} + \frac{5!}{3!} + \frac{4!}{2!} = 6 + 20 + 12 = 38$$

ways. The total is $89 + 38 + 1 = 128$.

Counting collections of coins instead of sequences gives 10, and ignoring the 50p coins gives 89.

Question 17. Answer **F**.

Only the remainder of the score on division by 3 matters. Doubling sends remainders $0, 1, 2$ to $0, 2, 1$, and adding 1 sends them to $1, 2, 0$. In particular, from remainder 1 both moves lead to remainder 2.

Track the probabilities of the remainders $(0, 1, 2)$ after each turn, starting from $(0, 1, 0)$. The new chance of remainder 0 is half the old chance of 0 (doubling) plus half the old chance of 2 (adding 1), and similarly for the others. This gives

$$(0,0,1),\ \left(\tfrac{1}{2},\tfrac{1}{2},0\right),\ \left(\tfrac{1}{4},\tfrac{1}{4},\tfrac{1}{2}\right),\ \left(\tfrac{3}{8},\tfrac{3}{8},\tfrac{1}{4}\right),\ \left(\tfrac{5}{16},\tfrac{5}{16},\tfrac{3}{8}\right),\ \left(\tfrac{11}{32},\tfrac{11}{32},\tfrac{5}{16}\right).$$

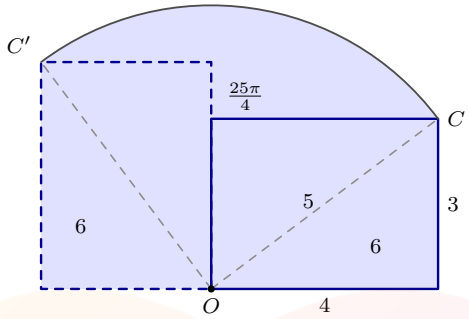
So the probability is $\dfrac{11}{32}$. The options $\dfrac{3}{8}$ and $\dfrac{5}{16}$ are the answers after four and five tosses, and $\dfrac{1}{3}$ is the guess that every remainder is equally likely.

Question 18. Answer **B**.

Let O be the fixed corner. The point of the rug furthest from O is the opposite corner C, at distance $\sqrt{4^2 + 3^2} = 5$, and the diagonal OC turns through 90°.

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Look along each ray from O. A ray lying behind the starting diagonal (on the side away from the turning) only meets the rug early on, and along it the rug reaches furthest at the start, because the edge of the rug moves inwards along the ray as the rug turns. So in those directions the covered region is the triangle cut off from the starting rug by its diagonal, of area $\frac{1}{2} \times 4 \times 3 = 6$. In the same way, rays beyond the finishing diagonal give a triangle of area 6 from the finishing rug. Every ray in between is crossed by OC at some moment, so it is covered out to distance 5 and no further, which gives a quarter disc of area $\frac{25\pi}{4}$.



The total is $12 + \dfrac{25\pi}{4} \approx 31.6$ m^2. The quarter disc alone misses the two triangles, and adding two whole rugs to it counts the parts inside the quarter disc twice.

Question 19. Answer **A**.

Because Cat's coin is fair, whatever Ana and Ben toss, Cat's coin matches any given face with probability $\frac{1}{2}$. So in each round:

if Ana and Ben agree, Cat is chosen or the round is replayed, each with probability $\frac{1}{2}$; if Ana and Ben disagree, Cat sides with one of them and the other is chosen.

Ana and Ben agree with probability $\frac{1}{3} \times \frac{1}{4} + \frac{2}{3} \times \frac{3}{4} = \frac{7}{12}$. So in a single round, Cat is chosen with probability $\frac{7}{24}$, and the round is replayed with probability $\frac{7}{24}$. Hence

$$P(\text{Cat is chosen}) = \frac{7/24}{1 - 7/24} = \frac{7}{17},$$

and Ana and Ben each have probability $\frac{5}{17}$, despite their different coins.

The option $\frac{7}{24}$ is the probability for a single round, forgetting the replays, and $\frac{1}{3}$ assumes the method is fair.

Question 20. Answer **D**.

Write $x = u_1$ and $y = u_2$, so that $y = x^2 - 2$ and $x = u_3 = y^2 - 2$. Subtracting these equations,

$$y - x = x^2 - y^2 = (x - y)(x + y),$$

so $(x-y)(x+y+1) = 0$. Since $y \neq x$, we need $x+y = -1$, that is $x + x^2 - 2 = -1$, or $x^2 + x - 1 = 0$. So $x = \dfrac{-1 \pm \sqrt{5}}{2}$; these two values are exchanged by the rule, since they add up to -1. The larger is $\dfrac{\sqrt{5}-1}{2} \approx 0.618$.

The values 2 and -1 also satisfy $u_3 = u_1$, but for them the sequence is constant, as $x^2 - 2 = x$; this case is excluded. Expanding $(x^2-2)^2 - 2 = x$ gives a quartic whose roots are 2, -1 and the two values above, but subtracting the two equations avoids having to solve it.

Question 21. Answer **E**.

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Each party's quotients $\frac{V}{1}, \frac{V}{2}, \frac{V}{3}, \ldots$ decrease, so awarding seats one at a time by largest quotient is the same as listing all the quotients of all the parties and giving one seat for each of the ten largest.

The quotients of A are $4200, 2100, 1400, 1050, 840, 700, \ldots$ and those of B are $3000, 1500, 1000, 750, \ldots$. Together, in decreasing order, they are

$$4200,\ 3000,\ 2100,\ 1500,\ 1400,\ 1050,\ 1000,\ 840,\ 750,\ 700,\ \ldots$$

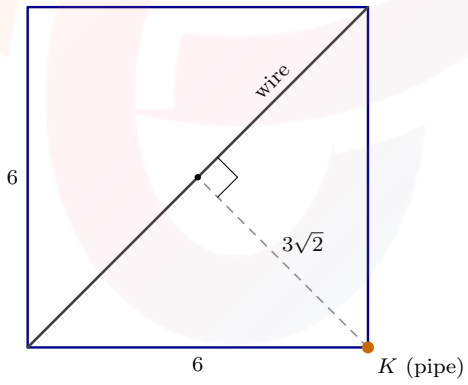
If C has V votes and wins 3 seats, then A and B win the other 7, for their seven largest quotients, down to 1000. So C's third quotient $\frac{V}{3}$ must beat the eighth value, 840, which belongs to the larger party A. This needs $\frac{V}{3} > 840$, so $V \geq 2521$, which is 721 extra votes; with $V = 2520$ the tie goes to A.

Requiring $\frac{V}{3} > 1000$ gives 1201, and asking for 30% of all the votes gives 1286.

Question 22. Answer **C**.

Look at the room from directly above. The pipe is vertical, so it appears as a single point K, a corner of the floor square, while the wire appears as the diagonal of the square that does not pass through K.

The distance between a point of the wire and a point of the pipe is at least the horizontal distance between them, and this is at least the distance from K to that diagonal, which is half of the other diagonal, $\frac{1}{2} \times 6\sqrt{2} = 3\sqrt{2}$. This value is attained: the foot of the perpendicular from K is the centre of the square, over which the wire passes at a height of 3 m, and the point of the pipe at a height of 3 m is level with it and $3\sqrt{2}$ away.



view from above

The option $2\sqrt{6}$ is the distance from either end of the pipe (a corner of the room) to the wire, which is not the shortest distance.

Question 23. Answer **B**.

Adding x changes the quartiles, so they must be recalculated for each case. An outlier must be the largest or the smallest of the eight times.

If $x \geq 50$, the ordered list is $30, 34, 35, 38$ and $41, 42, 50, x$, with quartiles $\frac{34+35}{2} = 34.5$ and $\frac{42+50}{2} = 46$. The interquartile range is 11.5, so x is an outlier when $x > 46 + 17.25 = 63.25$.

If $x \leq 30$, the list is $x, 30, 34, 35$ and $38, 41, 42, 50$, with quartiles 32 and 41.5. The interquartile range is 9.5, so x is an outlier when $x < 32 - 14.25 = 17.75$.

If $30 < x < 50$, the lower quartile is at most 34.5 and the upper quartile at least 41.5, so the interquartile range is at least 7, and the limits for outliers lie below 24 and above 52; so x is not an outlier.

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Hence x is not an outlier exactly when $18 \le x \le 63$, which is 46 values. Using the quartiles of the original seven times, 34 and 42, gives $22 \le x \le 54$, only 33 values.

Question 24. Answer **F**.

Count the $6^4 = 1296$ outcomes in which no two scores add up to 7. The scores that add up to 7 form the pairs $\{1,6\}$, $\{2,5\}$ and $\{3,4\}$, so the set of scores that appear must contain at most one number from each pair.

If only one number appears, there are 6 outcomes. If exactly two different numbers appear, they come from two different pairs, which can be chosen in $3 \times 2 \times 2 = 12$ ways, and each such set gives $2^4 - 2 = 14$ outcomes, 168 in all. If three different numbers appear, there is one from each pair, chosen in $2^3 = 8$ ways, and the outcomes using all three numbers number $3^4 - 3 \times 2^4 + 3 = 36$, giving 288.

So $6 + 168 + 288 = 462$ outcomes have no pair adding up to 7, and the required probability is $1 - \dfrac{462}{1296} = \dfrac{834}{1296} = \dfrac{139}{216}$.

The option $\dfrac{91}{216}$ comes from claiming that each die after the first has 5 allowed scores, which ignores the fact that the forbidden scores depend on which numbers have already appeared, not on how many dice have been rolled.

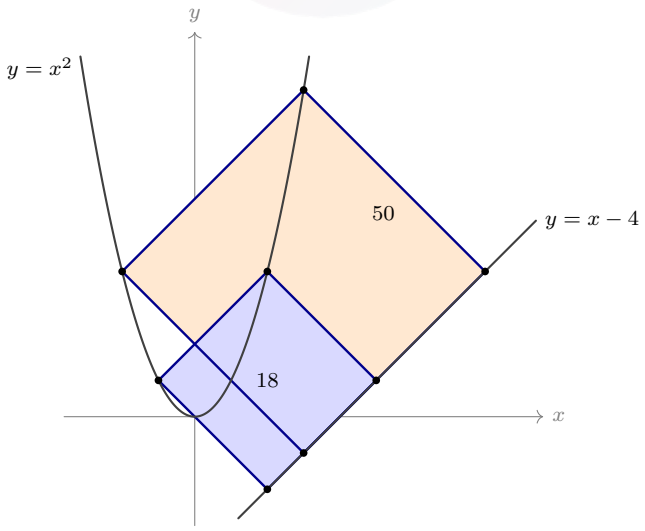
Question 25. Answer **C**.

The curve lies above the line, since $x^2 - x + 4 = \left(x - \frac{1}{2}\right)^2 + \frac{15}{4} > 0$. The two vertices on the line cannot be opposite corners of the square: if they were, the other two vertices would be reflections of each other in the line, so their midpoint would lie on the line, but the midpoint of a chord of $y = x^2$ lies above the curve. So the vertices on the line are adjacent, the side joining them lies along $y = x - 4$, and the opposite side lies along a parallel line $y = x + c$.

This line meets the curve where $x^2 - x - c = 0$; the roots x_1, x_2 satisfy $(x_1 - x_2)^2 = 1 + 4c$, and the side of the square on this line has length $\sqrt{2}\,|x_1 - x_2|$. Moving from this side to the opposite one is a shift by $(s, -s)$ with $s > 0$, which lowers $y - x$ by $2s$, so $c - 2s = -4$ and $s = \frac{c+4}{2}$; the side of the square is $s\sqrt{2}$. Hence

$$1 + 4c = \frac{(c+4)^2}{4}, \quad \text{so} \quad c^2 - 8c + 12 = 0,$$

giving $c = 2$ or $c = 6$.



The area is $2(1 + 4c)$: 18 when $c = 2$ (vertices $(-1,1)$, $(2,4)$, $(5,1)$, $(2,-2)$) and 50 when $c = 6$ (vertices $(-2,4)$, $(3,9)$, $(8,4)$, $(3,-1)$). The sum is 68. The two vertices on the line cannot instead

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be opposite corners: the other two corners would then be reflections of each other in $y = x - 4$, and reflecting (a, a^2) in that line gives $(a^2+4,\ a-4)$, which never lies on $y = x^2$ because $(a^2+4)^2 > a-4$ for every a. Forgetting the factor $\sqrt{2}$ in the side lengths gives 34, and finding only one square gives 18 or 50.

Question 26. Answer **F**.

Number the children 1 to 5 from shortest to tallest, and let $a_1, \ldots, a_5$ be the numbers along the row. In a jagged row the heights go up and down alternately, so either $a_1 < a_2 > a_3 < a_4 > a_5$ or $a_1 > a_2 < a_3 > a_4 < a_5$. Replacing each number h by $6 - h$ swaps these two patterns, so they occur equally often, and it is enough to count the first.

In the first pattern, child 5 cannot be in position 1, 3 or 5, as each of those children is shorter than a neighbour. If child 5 is in position 2, both of its neighbours are automatically shorter; choose a_1 from the other four children in 4 ways, and then the remaining three must satisfy $a_3 < a_4 > a_5$, so the tallest of them goes in position 4 and the other two can be placed in 2 ways. That gives 8 rows, and by symmetry there are another 8 with child 5 in position 4.

So there are $2 \times 16 = 32$ jagged rows out of $5! = 120$, and the probability is $\dfrac{32}{120} = \dfrac{4}{15}$. Counting only one of the two patterns gives $\dfrac{2}{15}$.

Question 27. Answer **E**.

Clearly $n = 1$ works. For $n > 1$, let P be the largest prime factor of n. Since P divides $f(n)$, which is a product of numbers $q + 1$ for primes q dividing n, the prime P divides some $q + 1$ with $q \leq P$. Now $q + 1 \leq P + 1$, and $q + 1 = P + 1$ is impossible because P does not divide $P + 1$; so $q + 1 = P$. Two consecutive whole numbers that are both prime must be 2 and 3, so $P = 3$ and $q = 2$.

Hence $n = 2^a 3^b$, and $f(n) = 3^a 4^b = 2^{2b} 3^a$. This is a multiple of n exactly when $a \leq 2b$ and $b \leq a$ (which forces $a, b \geq 1$ unless $n = 1$). Listing the possibilities up to 1000: $b = 1$ gives $6, 12$; $b = 2$ gives $36, 72, 144$; $b = 3$ gives $216, 432, 864$ (the next, 1728, is too big); and $b \geq 4$ gives numbers at least $2^4 \times 3^4 = 1296$.

So there are $1 + 2 + 3 + 3 = 9$ such numbers. Forgetting $n = 1$ gives 8.

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